MILITARY SPECIFICATION

ALUMINUM ALLOY 7475, ROD

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging of Aluminum Alloy 7475, Rod for use in the manufacture of cartridge cases.

1.2 Classification. The material defined in this specification shall be fabricated rod having a diameter up to 2.00 inches. The material shall be supplied in annealed "0" temper or the as fabricated "F temper as specified in the contract or purchase order. Other types of material form may be provided, subjected to prior approval of the contracting office.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standard (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-6088	-	Heat Treatment of Aluminum Alloys
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods
FED-STD-245	-	Tolerances for Aluminum and Magnesium Wrought Products

FSC 9530

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:05Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-649	-	Preparation for Shipment and Storage of Aluminum and Magnesium Products

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD, adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable:

ASTM B557-73	-	Tension Testing of Wrought Aluminum and Magnesium Products
ASTM E10-73	-	Brinell Hardness of Metallic Materials
ASTM E18-74	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 General material requirements.

3.1.1 Chemical composition. The chemical composition of the material shall conform to the limits shown below:

Silicon	- 0.10 maximum	Chromium	- 0.18-0.25
Iron	- 0.12 maximum	Zinc	- 5.2-6.2
Copper	- 1.2-1.9	Titanium	- 0.06 maximum
Manganese	- 0.06 maximum	Others, Each	- 0.05 maximum
Magnesium	- 1.9-2.6	Others, Total	- .015 maximum

3.1.2 Product characteristics.

3.1.2.1 Condition. The fabricated rod material shall be supplied with a standard mill finish. The rods may be lightly oiled provided the oil is readily removable by commonly used alkaline cleaners or degreasing methods. Where "F" temper is specified, the material shall be as uniform in mechanical properties as possible consistent with recognized industry processing techniques.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:05Z
Check the source to verify that this is the current version before use.

3.1.2.2 Dimensions and tolerances. Rod dimensions including straightness shall be as specified in the purchase order. Unless otherwise specified tolerances shall be in accordance with standard tolerances as specified in Federal Standard No. 245.

3.1.3 Mechanical characteristics.

3.1.3.1 Mechanical and hardness properties. Material of "0" temper shall conform to the mechanical property requirements of Table I. Material of "0" or "F" temper heat treated to T62 temper shall also conform to the mechanical property requirement of Table I. Samples of "0" or "F" temper material shall be heat treated to T62 temper in accordance with with practices outlined in MIL-H-6088 for 7075-T6.

Because of the relative ease of conducting a hardness test, values for this parameter are included in Table I. In the case of "0" or "T62" temper they are intended as a guide which must be verified by mechanical property testing to determine acceptability. In the case of "F" temper the action to be taken when values exceed the maximum shall be defined in the cotract or purchase order.

TABLE I. Mechanical characteristics.

Mechanical Properties			
Temper	"F"	"0"	T62
Diameter (inches)	Through 2.0	Through 2.0	Through 2.0
Tensile Strength (KSI)	---	40,000 max.	75,000 min.
Yield Strength (KSI)	---	24,000 max.	66,000 min.
Elongation (% in 2")	---	10 min.	7 min.
Hardness			
Hardness R_E	75 max.	63 max.	---
Hardness R_B	---	---	85 min.
Hardness HB 10/500/30	---	55 max.	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:05Z
Check the source to verify that this is the current version before use.

3.1.4 Stability. The annealed rod material shall conform to the mechanical properties for "0" material defined in 3.1.3.1, and shall remain stable and not exceed the specified maximum strength requirements during storage at ambient temperatures not exceeding 125°F.

3.1.5 Workmanship.

3.1.5.1 Internal soundness. The material shall be ultrasonic quality and free of internal defects such as pipes, inclusions and laminations.

3.1.5.2 Surface condition. The surface of the fabricated rod shall be free from slivers, seams, scratches, inclusions, stains, graphite, run out marks, dirty residue or oils other than permitted in 3.1.2.1 and other defects which will result in defective cartridge cases.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (Table II).

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2.

4.3.2 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall meet the specification requirements. The contractors inspection shall be subjected to verification by Government representative.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:05Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-A-63547(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Aluminum Alloy, 7475, Rod				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Chemical composition	1	Acc-0 Rej-1	3.1.1 4.4.5.1	4.5.3
	Mechanical properties "0" temper	1	Acc-0 Rej-1	3.1.3.1 4.4.5.2	4.5.4
	Mechanical properties heat treated	1	Acc-0 Rej-1	3.1.3.1 4.4.5.2	4.5.4
	Hardness	1	Acc-0 Rej-1	3.1.3.1 4.4.5.3	4.5.5
	Stability	1	Acc-0 Rej-1	3.1.4	4.5.6

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:05Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
4.4.2.1	Aluminum Alloy 7475, Rod				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Chemical composition	1	4.4.3 4.4.4	3.1.1	4.5.3
	Mechanical properties	1	4.4.3 4.4.4	3.1.3.1	4.5.4
	Hardness	1	4.4.3 4.4.4	3.1.3.1	4.5.5

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:05Z
Check the source to verify that this is the current version before use.

4.3.3.1 First article sample failure. Failure of the sample to comply with the requirements of the specifications shall result in sample disapproval.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. The lists and inspection defined herein are intended to verify Quality Conformance Inspection with the requirements defined in this specification.

4.4.3 Rejection. Any lot which fails to meet the requirements of this specification shall be rejected. When the rejectable feature can be easily identified a sorting procedure can be used to save a portion of the rejected lot. Any such sorting procedure must be defined, including details of acceptance and rejection, and agreed to by both supplier and user. Where the defect may be detrimental to the function of the end item, the established sorting procedure shall also require the approval of the contractor.

4.4.4 Retests. Lots which fail to meet the mechanical property requirements of this specification may be retested by testing two (2) specimens for each specimen which failed. Both of these retest specimens must meet the applicable requirements of Table I or the lot shall be rejected. In the case of "0" temper material, if the original failure to meet the properties is determined to be inadequate annealing, the entire lot may be reannealed and resubmitted for test. If the failure to meet the properties is determined to be improper heat treating of the test samples, another set of samples may be obtained and subjected to the test. Test results shall be subjected to verification by the Government representatives.

4.4.5 Testing.

4.4.5.1 Chemical composition. Chemical analysis shall be made in accordance with either Method 111.2 or Method 112.2 of Federal Test Method No. 151. At least one sample shall be taken from each group of ingots poured simultaneously from the same source of molten metal. Analysis shall be made for all elements specifically mentioned in 3.1.1. When compliance with the above cannot be established, at least one sample shall be taken from each 4000 pounds of fabricated rod in a lot and analyzed to determine compliance withthe requirements of 3.1.1.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:05Z
Check the source to verify that this is the current version before use.

4.4.5.2 Mechanical properties. Testing shall be in accordance with ASTM B557, latest issue, including size, location and shape of specimen. Mechanical property tests shall be made for 1000 feet or less of rod in each lot. At least one test shall be made in the "0" temper if material is delivered in the "0" temper and at least one test shall be made on a sample of material heat treated to the T62 temper from the "0" or "F" temper. A lot for the purpose of tensile testing shall consist of all fabricated rods of the same nominal diameter traceable to an annealing lot or lots when supplied in the "0" temper, or to an identifiable quantity of fabricated rod of the same nominal diameter produced in the same production run when supplied in the "F" temper, and submitted for inspection at the same time.

4.4.5.3 Hardness. When specified in the contract or purchase order, the material shall be hardness tested in accordance with the procedures of ASTM E10 or ASTM E18. These tests shall be conducted once for each 1000 feet or less of rod shipment of material. The frequency of additional hardness testing shall be at the discretion of the contractor. Annealed "0" temper material and T62 temper material which fails to meet the hardness requirement, shall not be rejected unless the material also to meet the hardness requirement. Procedures to follow if "F" temper material exceeds maximum hardness shall be as defined in the contract or purchase order.

4.4.5.4 Workmanship. The material shall be examined for conformance with the criteria defined in 3.1.5 in accordance with standards established jointly by representatives of the supplier and the case manufacturer.

4.4.6 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Test methods and procedures.

4.5.1 Internal soundness. The contractor shall conduct tests to verify that the material meets the requirements of 3.1.5.1. The test procedure acceptance/rejection standards criteria, material sample size, and frequency of examination shall be developed by the contractor and submitted to the Contracting Officer for approval.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:05Z
Check the source to verify that this is the currentversion before use.

4.5.2 Surface condition. The surface quality of the material shall be inspected using accepted commercial methods. The material supplier and the contractor shall jointly develop a set of visual standards which clearly define allowable limits for those defects which cannot be defined in writing.

4.5.3 Chemical composition. Chemical analysis shall be made in accordance with either Method 111.2 or Method 112.2 of Federal Test Method No. 151. The analysis shall be made to verify compliance with the elements specifically defined in 3.1.1.

4.5.4 Mechanical properties. A sample quantity of "0" temper material and a sample of "0" or "F" temper material heat treated to T62 temper in accordance with the practices outlined in MIL-H-6088 for 7075-T6 shall be mechanically tested in accordance with ASTM B557-73, including size, location and shape of the specimen. The test results shall conform to the tensile, yield and elongation requirements of 3.1.3.1.

4.5.5 Hardness. A sample quantity of material of "0" or "F" temper shall be hardness tested. The tests shall be conducted in accordance with the procedures outlines in ASTM E10 and ASTM E18.

4.5.6 Stability. A sample quantity of "0" material, when packaged in accordance with 5.1 shall be placed in a storage area with temperatures limited to a range of 60° to 125°F for a period of 120 days. Following exposure, sample specimens of the material shall be subjected to the mechanical properties test in 4.1.2.3.2.

5. PACKAGING

5.1 Fabricated rod shall be packed in accordance with the requirements of MIL-STD-649, latest issue, to Level C requirements. Maximum package weight shall be specified in the contract or purchase order.

5.2 Packaging. Fabricated rods shall be shipped one lot per package or box, except that when necessary to add material from a second lot to complete a standard package the two lots must be physically separated and identified to insure that they will not become mixed or reversed.

5.3 Marking. Each package shall be marked with the following information:

a. Supplier's name or trade mark.

b. Alloy and temper.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:05Z
Check the source to verify that this is the current version before use.

c. Size.

d. Lot number.

e. Gross and net weights.

f. Specification number.

5.4 Reports. Each production shipment as defined by the contractor shall be accompanied by three (3) copies of a report certifying that the material contained in the shipment has been manufactured and inspected in accordance with the requirements of 4.2 and has been found to conform. The detailed test data shall be on file for inspection by the material user.

Reports covering sample preproduction shipments shall include but not be limited to data comparing specification values of 3.1.1 and 3.1.3.1 with test data. Reports for preproduction lots shall also include specific test data for the provisions of 3.1.4 when "0" temper material is supplied. The procurement document shall clearly define what is required on the certification (see 6.2).

6. NOTES

6.1 Intended use. The aluminum rod covered by this specification is intended for use in the manufacture of 30MM cartridge cases.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

6.4 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment design as required to: Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. Request letter of submittal, state contractor, contractor number, specification number, item nomenclature and classification of defects or test paragraph.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 9530-A011)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Research
& Development Command
ATTN: DRDAR-QAF
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-A-63547 ALUMINUM ALLOY 7475, ROD

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:05Z
Check the source to verify that this is the current version before use.